

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (AIML) (Credit System) Degree Examinations**

January 2023

20AM501 – INTERNET OF THINGS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full questions** choosing **Two full questions** from **Unit – I & Unit – II each** and **One full question** from **Unit – III**.

Unit – I

	Marks	BT*	CO*	PO*
1. a) Write a note on "IoT impact in real world".	8	L*2	1	1
b) Discuss IoT challenges.	8	L2	1	1
c) What does IoT and digitization mean? Elaborate on this concept.	4	L2	1	1
2. a) Describe IoT World Forum (IoTWF) standardized architecture.	8	L2	1	1
b) With a neat diagram explain the simplified IoT architecture.	8	L2	1	1
c) Write a note on fog computing.	4	L2	1	1
3. a) Explain the different sensor types with examples.	8	L2	2	1
b) Define smart objects. Explain its characteristics.	6	L2	2	1
c) What are the advantages and disadvantages of sensor network? Explain the design constraints of wireless sensor networks.	6	L2	2	1

Unit – II

4. a) What is Zigbee? Explain 802.15.4 Physical layer, MAC layer and security.	8	L2	3	1
b) Explain LoRaWAN standard and alliance, MAC layer and security.	6	L2	3	1
c) List and explain the different communication criteria for IoT.	6	L2	3	1
5. a) With a neat diagram explain 6LOWPAN protocol header compression and fragmentation.	6	L2	4	1
b) List and explain the key advantages of internet protocol.	6	L2	4	1
c) Explain the 6TiSCH forwarding models.	4	L2	4	1
d) Explain the main factors applicable to IPV4 and IPV6.	4	L2	4	1
6. a) Explain tunneling legacy SCADA over IP networks and SCADA protocol translation with a neat diagram.	10	L2	4	1
b) Describe MQTT framework, message format and working in detail.	10	L2	4	1

Unit – III

7. a) Discuss the benefits of flow analytics.	6	L2	5	1
b) Describe the components of Flexible Net Flow architecture.	6	L2	5	1
c) Explain the types of data analysis results.	4	L2	5	1
d) Explain the elements of Hadoop with a neat diagram.	4	L2	5	1
8. a) Explain the different steps and phases of OCTAVE Allegro Methodology.	8	L2	5	1
b) Discuss the common challenges of OT security.	6	L2	5	1
c) Explain the different zones and levels for IoT control hierarchy with diagram.	6	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
